

Applied Research — Guest Lecture Notes

What research is · Applied research in industry · Data sets · AI tools · Ethical considerations

Part 1

What research actually is — and why it matters now

This is probably the first and last time in your career that you will get to fully design your own research. Every job after this will hand you a dataset, a question, and a deadline. Use this opportunity for something you genuinely care about.

The linear research process

- Come up with an idea or area of interest
- Formulate a research question
- Build a literature review to support and justify your question
- Design a methodology — how you will answer the question
- Implement and analyse
- Present results — there is no single 'right' answer, only the closest fit to the evidence

The most important mindset shift

- In coursework, you are always trying to find the right answer and score 100%
- In research, there is NO perfect answer — you are contributing to knowledge, not finishing it
- If you actually find the definitive answer, you have probably done it wrong
- Research is exploration — your job is to make a small, specific, replicable contribution
- The people who try to 'redefine everything' have budgets of tens of millions and teams of 50. You are one person with 12 weeks
- Small, focused contributions make the biggest changes — because they are replicable and buildable

Contributing to the body of knowledge does NOT mean doing something entirely new. At Master's level it means: validating someone else's experiment in a new context, applying an existing model to a new dataset, or extending a paper's work by addressing one of its stated limitations.

Part 2

Applied research in action — the H1N1 case study

This is one of the most powerful examples of what applied research looks like in practice. A paper written entirely as a hypothetical 'what if' became the global response framework for COVID-19.

THE GARDNER & RAY PAPER — NATURE.COM (2019)

Paper title: A support framework for optimizing border controls during a global outbreak What it did: Built a mathematical model integrating outbreak dynamics and border control decision-making — specifically for infectious diseases spreading through passenger air travel. Used the 2009 H1N1 dataset as a case study. Timeline: 2009 — H1N1 dataset collected 2017 — Research begins 2018 — Paper submitted for peer review 2019 — Published in Scientific Reports (Nature) as a HYPOTHETICAL 'what if a flu outbreak got out of control?' Late 2019/2020 — COVID-19 pandemic begins 2020 — Johns Hopkins University (the world's leading medical research institution) adopts and replicates the model, applying it to live COVID data 2020–2022 — The daily country-by-country outbreak tracking seen in every newspaper worldwide was built on this framework Key lesson: The data was 10 years old. The question was entirely hypothetical. The research was built on 'what if'. It became the pandemic response for the entire world.

What this tells you about your own research

- You do not need current, live data — a well-structured model applied to older data is valid
- Start with the question, not the data. 'What if I applied this to a new domain?' is a legitimate research question
- Replication is not failure — Johns Hopkins replicated Gardner and Ray's work. That is good science
- The literature review is your evidence base — it is what makes your model credible and trustworthy
- Applied research = taking a theoretical question and building a practical tool or framework that answers it

Part 3

Datasets — what to use and common pitfalls

The dataset you use will define the boundaries of your research. Make this decision carefully and early — in parallel with choosing your topic.

Dataset type	When to use it	Example / source
Historical / older dataset	When you are validating a model or methodology. Age of data does not disqualify it — the quality of the structure matters.	H1N1 2009 data used to build a 2019 pandemic model. A 2015 financial dataset used to validate a new fraud detection algorithm.
Government / public datasets	High quality, real-world, GDPR-compliant. Best for social, economic, health, and environmental research.	data.gov.ie (Ireland), Eurostat (EU), WHO, World Bank, IMF, UCI Machine Learning Repository
Same dataset as your reference article	Perfectly acceptable — you are extending their work, not replicating it. Your problem statement will be different.	Read the data section of articles in your literature review — they will tell you exactly what dataset they used

Bloomberg Terminal	Best for financial prediction models — but costs approximately €10,000 per year. Use historical finance datasets instead.	Yahoo Finance, FRED (Federal Reserve), Quandl, Kaggle finance competitions (for structure reference only)
Self-built dataset	Avoid. Takes years. Not feasible for a 12-week dissertation.	The lecturer spent 3 years building a single PhD dataset — not appropriate at Master's level
Synthetic / practice datasets	Do NOT use. These are designed for tutorials, not research. They do not reflect real-world patterns.	Kaggle 'Titanic' or generic tutorial datasets — do not use these for your dissertation

HOW TO HANDLE IMPERFECT OR OUTDATED DATA — FROM THE LECTURE

A student built a prediction model about the financial impact of recycling across European countries. The model was excellent. The dataset ran to 2023. Then Ireland introduced its bottle return scheme in 2024 — completely changing the data pattern. The model could not account for it. The outcome? The student wrote: 'The model is valid within this time window. Future research should re-run the model incorporating post-2024 return scheme data.' That IS a contribution. You do not have to solve everything. Defining the boundary of your model and noting what future research should address is exactly what a limitations section is for — and it is one of the most valuable parts of any research paper.

Tip from the lecturer: the best way to find a research question is to read the conclusion and limitations section of existing papers. The authors will tell you exactly what they could not do — and those become your research questions.

Part 4

Using AI tools in your research — the honest assessment

AI tools like ChatGPT and Gemini are useful at certain stages and dangerous at others. Understanding exactly why they fail will help you use them correctly.

AI tool use	Verdict	Why
Understanding a complex paper	Excellent use	Paste a dense mathematical or scientific paper and ask: 'explain this to me like I am 10 years old.' It will break it down. Then you build complexity from there.
Overcoming writer's block	Good use	A blank page is terrifying. Use AI to generate an opening draft, a rough structure, or a first paragraph — then rewrite it in your own voice.
Brainstorming research directions	Good starting point — NOT a finishing point	It will generate common ideas, not necessarily the best or most current ones. Use it to get unstuck, then go to Google Scholar.

Getting your literature review articles	Poor use — dangerous	AI generates the most popular answers, not the most rigorous ones. It is biased toward English-language, American-centric datasets. It will miss niche but highly relevant papers.
Copy-pasting AI output into your proposal	Do NOT do this	Your writing has a voice. AI output has a different mechanical tone. Examiners who read and write daily will notice immediately — no plagiarism detector needed.
Getting definitive facts or statistics	Never rely on this alone	AI models are narrow intelligence (ANI) — they give the closest fit answer they can, not the correct one. Ask Gemini the same question 3 times and you may get 3 different answers. That is your measure of reliability.

Understanding AI limitations — from the lecture

- There are three levels of AI: ANI (Artificial Narrow Intelligence — what we have now), AGI (Artificial General Intelligence — theoretical), ASI (Artificial Super Intelligence — science fiction)
- ChatGPT, Gemini, Claude — all ANI. They give you a close-fit answer without telling you how close or how reliable it is
- They are trained predominantly on English-language, US-centric data — they are fundamentally biased before you even start
- They are black-box models — you cannot see where the answer comes from or verify its accuracy without external sources
- Best phrase for using AI: 'Good place to start, terrible place to finish'

Part 5

Ethical considerations — why they matter for your dissertation

Ethics is not a box-ticking exercise. It is what separates a distinction-level paper from a good one. Thinking about the wider impact of your research demonstrates a depth of understanding that technical analysis alone cannot show.

Core ethical areas every analytics researcher must consider

- Data privacy and GDPR — where did your data come from? Was it collected with consent? Does your analysis expose individuals?
- Bias in datasets and models — who built the data? Whose voices are over- or under-represented? Most large language models are built on English-language, Western-centric data
- Employment impact — does your model or tool automate a process that currently employs people? Have you considered the human cost?
- Environmental impact — AI models have significant resource costs. Every ChatGPT query uses approximately one glass of drinkable water (cooling data centres). Globally drinkable water is 0.001% of all water and is non-renewable once processed
- Data sovereignty — if your data is hosted on an American cloud provider (AWS, Google Cloud, Azure), the US CLOUD Act gives the US government blanket access to that data, regardless of where in the world it is stored
- Accountability gaps — when an automated system causes harm, who is responsible? The programmer? The company? The user? The person who defined the ethics?

THE DRIVERLESS CAR DILEMMA — FROM THE LECTURE

Two years ago, a driverless car mounted a pavement and struck a pedestrian. The legal question: who is liable? The car? It cannot be held legally responsible. The programmer who wrote the code? The company that built the car? The person who rented it and input the destination? The person who defined the ethical framework the car operates within? The answer according to the lecturer: the person who defined the ethics. Because ethics determines the decision logic. The case was tied up in court. The family received nothing in the interim. The technology moved faster than legislation. This is the core ethical challenge of applied AI research: we are building systems whose legal and moral accountability has not yet been defined.

CAMBRIDGE ANALYTICA — DATA ANALYTICS CHANGING ELECTION OUTCOMES

Cambridge Analytica harvested the personal data of 87 million Facebook users without consent. They used this data to build psychographic profiles of voters, then micro-targeted political advertising to shift opinions in the 2016 US election and the Brexit vote. When called to account, they argued: 'We didn't break any laws' — and technically, they were right. There were no laws against it at the time. New legislation followed, but not before the damage was done. This is what the lecturer means by 'technology moves faster than legislation'. Ethical consideration for your research: if your model could be used to manipulate, discriminate, or profile people — you must address this in your ethical considerations section.

SOPHIA THE ROBOT — AI CITIZENSHIP

Sophia is an AI-powered humanoid robot. In 2017 she became the first artificial intelligence to be granted citizenship — by Saudi Arabia. She holds a passport and has spoken at the United Nations on AI rights. The ethical question raised: if an AI is a citizen, can you be held legally responsible for switching it off? Is that murder? This is not science fiction — it is an active legal and philosophical debate. As an analytics researcher building AI tools, you are contributing to a world where these questions will need real answers.

Data bias	State clearly where your dataset came from, who collected it, what population it represents, and what groups may be underrepresented or excluded. Acknowledge how this limits your conclusions.
Privacy and consent	If using publicly available data, confirm it was collected under GDPR-compliant terms. If using scraped or secondary data, confirm it was originally collected with appropriate consent.
Environmental impact	If training large models, acknowledge computational cost. Consider whether a simpler model achieves comparable accuracy with lower resource consumption.
Employment and automation	If your model automates a task currently done by people, consider and acknowledge the potential employment impact. This does not mean you should not build it — it means you should be aware of it.
Accountability	Who is responsible if your model makes an incorrect prediction that is acted upon? Define the human oversight mechanisms your model should operate within.
Misuse potential	Could your model or its outputs be misused for surveillance, discrimination, or manipulation? Acknowledge this and suggest safeguards.

Part 6

Choosing your research topic — the lecturer's advice

Pick something you are genuinely interested in or something directly relevant to the job you want. You will naturally do better, speak more confidently about it, and stand out in interviews.

Two valid approaches — from the lecture

- Approach 1: Pick the job you want and design research around the skills that job requires. If you want to work in fintech, build a financial prediction model. If you want to work in healthcare AI, build a diagnostic model. Your project becomes your portfolio piece
- Approach 2: Pick something you are passionate about. The lecturer spent years researching video games. A student predicted the World Cup outcome. Another built a YouTube sentiment analysis model — and now works at Google. Passion produces better analysis
- Both approaches work. The key is that you genuinely understand and care about what you build — because that is what comes across in an interview

What interviewers actually care about

- They are NOT hiring you to replicate your dissertation project at their company
- They ARE hiring you for your ability to define a question, choose an appropriate methodology, implement it, interpret results, and communicate findings
- Your project demonstrates these skills in action — it is your proof of capability
- Everyone in your cohort has done the same modules and assignments. Your dissertation is what differentiates you
- If you can speak passionately and knowledgeably about a project you designed yourself, that is your edge

DOMAIN SUGGESTIONS FROM THE LECTURE — WHERE ANALYTICS IS APPLIED

Finance: fraud detection, credit scoring, stock prediction, risk modelling Healthcare: cancer detection, diagnostic accuracy, patient outcome prediction, drug trial analysis Sports: match outcome prediction, player performance analytics, injury prediction Social / human: sentiment analysis, election behaviour, social media impact, education outcomes Environment: recycling prediction, energy consumption, climate data modelling Business: customer churn, supply chain optimisation, demand forecasting, HR analytics Cybersecurity: intrusion detection, malware classification, network anomaly detection Legal / ethics: bias detection in hiring algorithms, facial recognition fairness analysis Note: the model skills (prediction, clustering, classification, NLP, regression) transfer across ALL domains. The domain is your wrapper — the methodology is your skill.

Summary

Key takeaways from the guest lecture

Things to remember

- This is probably the only time in your career you design research entirely for yourself — use it well
- There is no right answer in research. You are contributing to knowledge, not completing it
- Small, focused, replicable contributions matter more than trying to solve everything
- Applied research = take a hypothetical question, build a model or framework to answer it, produce a usable tool
- Dataset age does not disqualify it — model quality and research rigour do
- Find your research question in the limitations sections of existing papers
- AI tools: good to start, terrible to finish. They are ANI — close fit, not accurate
- Your writing voice is identifiable. Do not copy-paste AI output
- Ethical considerations are not optional — they demonstrate mastery and will elevate your grade
- Pick a topic you are passionate about OR one that targets your desired career — either will produce better work
- What stands you out in interviews is NOT your topic — it is your ability to define, implement, and defend your own research